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# Pharmacotherapy of allergic rhinitis

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## INTRODUCTION

Allergic rhinitis is characterized by sneezing, rhinorrhea, and nasal obstruction as well as pruritus of the nose and palate. It is frequently associated with postnasal drip, cough, irritability, and fatigue, as well as itchy, watery eyes in those with concomitant allergic conjunctivitis. Allergic rhinitis is commonly associated with comorbid asthma. Symptoms can be present year-round, seasonally, or episodically upon exposure to specific allergens.

The pharmacologic management of allergic rhinitis is presented in this topic. Our recommendations are consistent with consensus guidelines [1-13].

The clinical manifestations, diagnosis, differential diagnosis, and pathogenesis of allergic rhinitis are discussed separately. (See "Allergic rhinitis: Clinical manifestations, epidemiology, pathogenesis, and diagnosis".)

Additional treatment strategies, including allergen avoidance, allergen immunotherapy, and complementary and alternative therapies, are also discussed separately:

- (See "Allergen avoidance in the treatment of asthma and allergic rhinitis".)
- (See "Subcutaneous immunotherapy (SCIT) for allergic rhinoconjunctivitis and asthma: Indications and efficacy".)
- (See "Sublingual immunotherapy for allergic rhinitis and conjunctivitis".)
- (See "Complementary and alternative therapies for allergic rhinitis and conjunctivitis".)

## NONPHARMACOLOGIC STRATEGIES FOR ALL PATIENTS

We advise allergen avoidance and nasal saline as nonpharmacologic therapy for all patients with allergic rhinitis. These strategies are helpful for any severity level and are advised regardless of the pharmacotherapy used. They may be particularly useful for patients with concerns about adverse effects from pharmacotherapy (eg, young children or pregnant individuals).

**Allergen avoidance** — We advise allergen avoidance (to the extent possible) for patients who have an obvious allergy based on clinical history and in those patients with documented immunoglobulin E (IgE) mediated allergic responses to specific allergens. Common allergens include pets, dust mites, pollen, and mold. Indications for allergen testing and specific guidance on allergen avoidance are discussed in detail elsewhere ( table 1). (See "Allergic rhinitis: Clinical manifestations, epidemiology, pathogenesis, and diagnosis", section on 'Allergen-specific testing' and "Allergen avoidance in the treatment of asthma and allergic rhinitis".)

**Nasal saline** — We advise nasal saline spray or irrigation for all patients with allergic rhinitis. For those with mild symptoms, saline alone can be helpful in improving symptoms. For patients with more severe symptoms who use additional pharmacologic nasal sprays, saline use prior to administration of other topical agents can help improve the efficacy of the pharmacologic spray.

Nasal saline spray or irrigation can be performed as needed, daily at baseline, or twice daily for increased symptoms. The mechanism of benefit is through washing of allergens from the nasal passages, as well as the soothing effect of the saline.

- **Optimal technique** – Optimal technique helps improve the efficacy of nasal irrigation. Detailed instructions for patients are provided ( table 2). Patients can make their own irrigation solutions or buy commercially prepared solutions or kits. The saline

solution may be warmed or used at room temperature.

- **Optimal timing** – For those also using pharmacologic nasal sprays, we advise saline use approximately 10 to 15 minutes prior to using the pharmacologic nasal spray. This allows the saline to drain completely from the nose and increases the likelihood that the medication will be absorbed as fully as possible.

- **Large-volume irrigation preferred** – We generally prefer large-volume irrigation (>200 mL per side) when possible because this is more effective in relieving symptoms compared with saline nasal sprays; however, large-volume irrigation may not always be convenient or practical. It is typically done with over-the-counter irrigation devices, such as squeeze bottles, neti pots, or bulb syringes. Large-volume irrigations should only be suggested to patients old enough to perform the irrigation themselves.

- **Adverse effects** – Nasal irrigation carries little risk if properly performed. Patients should use distilled, sterilized, or previously boiled water to prevent rare but fatal amebic meningoencephalitis infections, which have been reported from use of contaminated tap water [14,15].

- **Efficacy** – Saline irrigation improves symptoms in a variety of rhinitis conditions [16-18]. In one randomized trial of children with acute sinusitis and concomitant allergic rhinitis, nasal irrigation significantly improved rhinorrhea, nasal congestion, throat itching, sleep quality symptoms, and nasal airflow [16]. Another study in children found that the effects of nasal irrigation were additive with those of intranasal glucocorticoids [17]. Nasal irrigation is particularly helpful when there are crusted nasal secretions due to chronic, thick drainage.

## ADULTS AND CHILDREN TWO YEARS AND OLDER

For adults and children ≥2 years of age, the approach to pharmacotherapy depends on the severity and persistence of symptoms as well as the presence of comorbid conditions. Classification of allergic rhinitis with symptom severity is shown in the table ( table 3).

In general, patients who self-treat with over-the-counter agents tend to overutilize older, sedating antihistamines and underutilize glucocorticoid nasal sprays ( table 4). The challenge for clinicians is to educate patients about the relative efficacy of different therapies and ensure that patients are adequately treated with medications that do not cause undue side effects.

**Mild or intermittent symptoms** — All pharmacotherapy is most effective when taken regularly, but as-needed use may be sufficient for patients with mild or intermittent symptoms that are related to predictable allergen exposures (eg, visiting a relative's house with a pet).

- **Glucocorticoid nasal spray** – We prefer initial therapy with a glucocorticoid nasal spray as this is more effective than oral antihistamines ( table 5) [19,20]. When used as needed for predictable exposures (eg, a visit to a house with pets), we advise initiating therapy two days before, continuing through, and for two days after the end of exposure [19,21]. For persistent symptoms or unpredictable exposures, glucocorticoid nasal sprays should be used regularly.

  In a meta-analysis of eight randomized trials, as-needed glucocorticoid nasal spray improved nasal symptom scores compared with placebo and as-needed oral antihistamines, but it was not as effective as regular use of glucocorticoid nasal spray [20].

  Additional considerations related to glucocorticoid nasal sprays, including optimal technique and adverse effects, are discussed below. (See 'Preferred therapies' below.)

- **Antihistamine nasal spray** – For patients who do not improve with glucocorticoid nasal sprays or prefer an alternative option, we use an antihistamine nasal spray (eg, azelastine or olopatadine). These are effective in symptom relief and can be taken immediately prior to exposure. Thus, they are a good option for patients who have unpredictable or unplanned exposures. (See 'Antihistamine-only nasal sprays' below.)

- **Combination glucocorticoid/antihistamine nasal spray** – Combination glucocorticoid/antihistamine nasal sprays are also considered first-line therapy. In the US, available formulations include azelastine-fluticasone (approved for those ≥6 years), and olopatadine-mometasone (approved for those ≥6 years), which require prescriptions. Their onset of action is typically within 15 to 30 minutes due to the rapid action of the antihistamine component. (See 'Combination glucocorticoid/antihistamine nasal spray' below.)

- **Minimally sedating oral antihistamines** – For patients who do not respond to glucocorticoid and/or antihistamine nasal spray and for those who prefer oral therapy, we advise using a minimally sedating oral antihistamine ( table 6). When used as needed, cetirizine, fexofenadine, rupatadine, and bilastine should be taken two to five hours before an exposure, whereas

loratadine is ideally taken approximately eight hours prior to exposure. These agents are similarly efficacious. (See 'Minimally sedating oral antihistamines' below.)

- **Cromolyn nasal spray** – Cromolyn nasal spray is an alternative option that can be administered as needed, ideally 30 minutes before an exposure. It is only effective as a prophylactic medication. Taken in this manner, cromolyn is helpful for brief exposures (minutes to hours). For prolonged exposures, administration should ideally begin four to seven days in advance. Cromolyn may be preferred for patients with heightened concerns about adverse effects (eg, children and pregnant individuals) because of its minimal bioavailability. However, it is not as effective as other therapies. (See 'Therapies with limited roles' below.)

**Seasonal symptoms** — We prefer glucocorticoid nasal sprays (with or without intranasal antihistamine) for seasonal allergy symptoms [10,12,13]. Antihistamine-only nasal sprays, minimally sedating oral antihistamines, and cromolyn nasal spray are alternatives for patients who do not respond to glucocorticoid nasal spray and/or have preferences for alternative agents (eg, desire for oral agent or concern about glucocorticoid safety).

For patients with predictable seasonal allergies, we advise initiating regular treatment at least a week (ideally longer) before the pollen season rather than after symptoms have begun. Medication use can be decreased as tolerated as pollen season ends. In the United States, the National Allergy Bureau is a useful resource for patients who would like information about pollen counts in their area. In Canada, Aerobiology Research Labs provides this service.

**Persistent or moderate-to-severe symptoms** — Most patients with persistent or moderate-to-severe symptoms can be treated with glucocorticoid nasal sprays or combination glucocorticoid/antihistamine nasal sprays (in older children and adults). Additional options include antihistamine-only nasal spray, minimally sedating oral antihistamines, and/or combined oral antihistamine/decongestants.

We have a low threshold for referring children ≥5 years old with moderate-to-severe symptoms (table 3) to an allergist for consideration of allergen immunotherapy as this can reduce their risk of disease progression and development of asthma. (See 'Children with moderate-to-severe or persistent symptoms' below.)

**Preferred therapies** — We recommend treatment with a glucocorticoid nasal spray for adults and children ≥2 years old with persistent or moderate-to-severe allergic rhinitis; combination glucocorticoid/antihistamine nasal sprays are an alternative first-line option for children ≥6 years old and adults.

**Glucocorticoid nasal spray** — Our approach is to start therapy with the maximal dose for age (table 5). Once symptoms are adequately controlled, the dose can be "stepped down" at one-week intervals to the lowest effective dose. Patients with severe symptoms will require daily use on a chronic basis. Some patients can reduce the frequency of use gradually and maintain symptom control with every-other-day or as-needed use.

- **Onset of action and efficacy** – Most glucocorticoid nasal sprays have an onset of action of a several hours [22-24]. However, maximal effect may require several days or weeks in patients with longstanding untreated symptoms. Glucocorticoid nasal sprays are the most effective single pharmacologic therapy for allergic rhinitis and are recommended by guidelines as the best single therapy for patients with persistent or moderate-to-severe symptoms [10,12]. Several meta-analyses have demonstrated greater reductions in nasal symptom scores with glucocorticoid nasal sprays compared with oral antihistamines [25,26]. These agents are particularly effective in the treatment of nasal congestion, which often does not respond to oral antihistamines [1,19,25,27]. Glucocorticoid nasal sprays have also been shown to be more effective than antihistamine nasal sprays in most studies [5,28].

- **Choice of spray** – We prefer minimally bioavailable nasal sprays with once-daily dosing when possible as these are more convenient, may help improve adherence, and may be safer for long-term use than agents with higher systemic bioavailability (table 5) [29-32]. We particularly prefer those glucocorticoid nasal sprays with low bioavailability and once-daily dosing in children as they may be less likely to cause side effects due to minimal systemic absorption, although this has not been proven. These preferred options include fluticasone furoate, fluticasone propionate, mometasone, and ciclesonide. All of the preparations are similarly effective [33,34].

  For patients who dislike the wet run-off or taste side effects of some aqueous sprays, products that use a "dry" aerosol delivery system may be preferred (table 5). In the United States, this includes beclomethasone dipropionate hydrofluoroalkane (HFA; Qnasl [brand name]) and ciclesonide HFA (Zetonna [brand name]).

  Drug interactions are an additional consideration. Specifically, fluticasone nasal sprays should be avoided in patients treated with strong inhibitors of CYP3A4 enzymes (eg, ritonavir, itraconazole, nefazodone). (See 'Safety and adverse effects' below.)

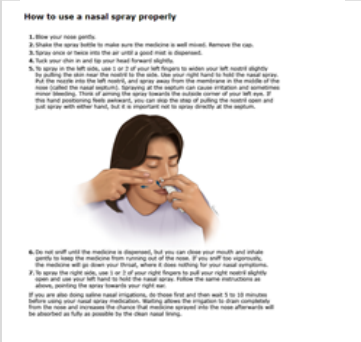
If a patient does not respond to one glucocorticoid nasal spray after consistent use, other formulations are not likely to be more effective. We add alternative agents rather than switching to a different nasal glucocorticoid. (See 'Additional options' below.)

**Combination glucocorticoid/antihistamine nasal spray** — In children ≥6 years old and adults, combination glucocorticoid/antihistamine nasal sprays can be used as first-line therapy or as second-line therapy in patients who do not improve with glucocorticoid-only nasal spray. We do not usually use antihistamine nasal sprays in younger patients, as they can be sedating.

- **Onset of action and efficacy –** The onset of action of glucocorticoid/antihistamine combination sprays is usually within 15 to 30 minutes (due to the rapid onset of action of the antihistamine component). Several studies have demonstrated the efficacy of a combination glucocorticoid/antihistamine nasal spray compared with either agent alone [35-37]. As an example, in one randomized trial of almost 3400 patients with moderate-to-severe allergic rhinitis, those who received the combination of azelastine-fluticasone were most likely to experience near-complete resolution of total symptoms after two weeks (12 percent of patients compared with 9 percent with fluticasone, 7 percent with azelastine, and 4 percent with placebo) [36].

- **Choice of spray –** In the United States, available formulations are azelastine-fluticasone and olopatadine-mometasone. Each is dosed twice daily. Azelastine-fluticasone is approved for those ≥6 years, whereas olopatadine-mometasone is approved for those ≥12 years (≥6 years in Canada). The agents have not been directly compared, although in our experience they are similarly efficacious. When combination sprays are not available, we instruct patients to add an antihistamine spray to the glucocorticoid spray they are using.

**Optimal technique and use** — To optimize the effects of nasal sprays, we suggest the following ( figure 1):




How to use a nasal spray properly

Figure 1 - larger image below

- **Optimal technique –** Proper positioning of the head can prevent the spray from draining down the throat. Specific guidance depends on the type of spray ( table 5):

  - With the aqueous (nonaerosol) nasal sprays, the patients should keep their head pointed slightly downward while spraying and avoid tilting the head back. In addition, they should avoid pointing the spray at the septum, which can become irritated.

  - With the aerosol products, patients should tilt their head back slightly, dispense the spray, hold their breath for a few seconds, and then exhale through their mouth. The product insert also advises to avoid blowing the nose for 15 minutes after use. The nosepiece should be wiped with a clean, dry tissue weekly, although not cleaned with water.

  A patient education topic on rhinitis, which reviews techniques for use of nasal sprays, is provided separately. (See "Patient education: Allergic rhinitis (Beyond the Basics)".)

- **Pretreatment with saline nasal spray or irrigation –** If mucous crusting is present, patients can rinse the nose with a saline nasal spray or irrigation before administering the nasal glucocorticoid. Mucus or other debris can prevent the medication from coating the nasal mucosa. (See 'Nasal saline' above.)

- **Pretreatment with decongestant spray –** For patients who are so obstructed that the spray will not penetrate into the nasal space, we advise use of a decongestant spray 10 minutes before the nasal glucocorticoid. This should only be done for three to five or even fewer days to avoid the complications of decongestant sprays. After that, the glucocorticoid spray should be continued alone. If the nasal decongestant spray is not effective in relieving the obstruction, a very short course of oral glucocorticoids may be used. (See 'Therapies with limited roles' below.)

**Safety and adverse effects** — Glucocorticoid nasal therapy (with or without antihistamine) is generally very safe. While there are some risks of long-term glucocorticoid use, these are small due to the relatively low doses involved and limited glucocorticoid steroid absorption(particularly with minimally bioavailable formulations). However, there does appear to be a small decrease in growth in children with long-term use. Additionally, local side effects such as irritation and epistaxis are common. Because adverse effects are possible, glucocorticoid nasal sprays of any type should be tapered to the lowest effective dose in all patients once symptoms are controlled. (See 'Optimal technique and use' above.)

- **Decreased growth in children with long-term use –** Growth in children should be monitored when any glucocorticoid-containing medication is prescribed, especially if additional glucocorticoid-based medications are given, such as inhaled agents for asthma or topical corticosteroids for atopic dermatitis [38]. In addition, as discussed above, once-daily minimally bioavailable sprays are preferred in growing children since they may be less likely to impact the hypothalamic-pituitary axis (HPA) and growth ( table 5) [29].

  Most studies have shown no or limited HPA suppression at recommended doses [39-45]. However, several studies have suggested a small effect on growth [46-49]. As an example, in one randomized trial of over 470 prepubescent children (ages 5 to 8.5 years at screening) with perennial allergic rhinitis, those who received fluticasone furoate for 52 weeks experienced a reduction in growth velocity of -0.27 cm/year compared with those who received placebo (95% CI -0.48 to -0.06 cm/year) [49]. However, the dose of active agent was 110 mcg daily, which is the maximum recommended dose for this age group, and the protocol used in this study did not allow for dose reduction once symptoms were controlled. There was no detectable change in 24-hour urinary cortisol excretion to suggest adrenal suppression.

  Longer-term studies of the effects on growth caused by inhaled glucocorticoids administered for asthma treatment (which deliver higher doses per actuation) are reassuring, although caution is still warranted. (See "Major side effects of inhaled glucocorticoids", section on 'Growth deceleration'.)

- **Other effects of long-term glucocorticoid absorption –** A small number of studies have examined the effects of glucocorticoid nasal sprays on other adverse effects related to long-term glucocorticoid absorption, such as decreased bone mineral density, increased intraocular pressure, or cataract formation [50-52]. A meta-analysis of 10 randomized, controlled trials comprised of 2226 adult and adolescent patients with allergic rhinitis did not identify a clear risk of elevated intraocular pressure or development of posterior subcapsular cataracts after treatment for one year with glucocorticoid nasal sprays [53]. Some studies have demonstrated detrimental effects, although it is unclear whether these are large enough to result in clinically important outcomes over time.

- **Drug interactions –** We advise that patients treated with strong inhibitors of CYP3A4 enzymes (eg, ritonavir, itraconazole, nefazodone) use a glucocorticoid spray **other** than fluticasone. A table of strong CYP3A4 inhibitors is provided ( table 7). Drug interactions between intranasal fluticasone and strong inhibitors of CYP3A4 enzymes can effectively raise the dose of glucocorticoid and result in clinically significant adrenal suppression [54]. Although it is unclear how frequently this adverse outcome occurs, avoiding fluticasone treatment in this group is reasonable. Budesonide, beclomethasone, triamcinolone, and flunisolide appear to be safer options. (See "Major adverse effects of systemic glucocorticoids", section on 'Organ-based toxicity of systemic glucocorticoids'.)

  For specific interactions, use the drug interactions program included with UpToDate.

- **Local adverse effects –** Local adverse effects include irritation, epistaxis, and rare nasal septal perforation.

  - **Local irritation –** Local irritation of the nasal mucosa, including drying, burning, and/or discomfort, is reported by 2 to 10 percent of patients using glucocorticoid sprays [55]. Formulations containing alcohol or propylene glycol are more irritating than aqueous preparations ( table 5). Sometimes patients can minimize these complications by using proper administration technique and by using the lowest effective dose of the agent. (See 'Optimal technique and use' above.)

  - **Epistaxis –** Scant bleeding due to mucosal irritation is common. It can often be remedied by stopping treatment on the affected side of the nose for a few days and then restarting therapy. Proper administration technique should be reviewed.

    Significant epistaxis is reported with both glucocorticoid nasal sprays and placebo in clinical trials and may therefore partly result from mechanical trauma of repeated spraying, although, when placebo rates were subtracted, nosebleed was still noted in 2 to 12 percent of patients using different glucocorticoid preparations. This can be particularly bothersome for older patients, who may do better with antihistamine spray if the problem persists. Frank epistaxis is often hard to prevent and, once it occurs, may require future avoidance of glucocorticoid nasal sprays. Recurrent or chronic epistaxis should prompt evaluation for chronic nasal inflammation. (See "Chronic rhinosinusitis: Clinical manifestations, pathophysiology, and diagnosis", section on 'Evaluation'.)

  - **Rare septal perforation –** Nasal sprays should always be directed away from the septum. There are rare reports of nasal septal perforation with glucocorticoid nasal sprays [56,57]. However, long-term studies of large numbers of patients have not found evidence of damage to the nasal mucosa despite years of use in the majority of patients [1,57].

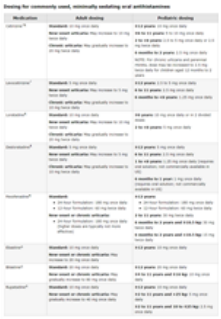
**Additional options** — For patients with moderate-to-severe symptoms who do not respond adequately to initial therapy with glucocorticoid nasal spray, a second agent can be added to their regimen. Options include an antihistamine nasal spray, oral

antihistamines, and oral antihistamine/decongestant combination products. There are few clinical trials directly comparing different combinations of these therapies, and the choice of additional agents should be based on the patient's residual symptoms, preferences, and comorbid conditions.

**Antihistamine-only nasal sprays** — Antihistamine nasal sprays (eg, azelastine and olopatadine) without glucocorticoid are an option for patients who prefer to avoid glucocorticoids. These agents appear to have some antiinflammatory effect and can improve nasal congestion [58-60]. They have a rapid onset of action (less than 15 minutes) and can be administered "on demand" [2,23].

Azelastine and olopatadine are similarly effective [61-63]. In noninferiority trials, both compared favorably with fluticasone propionate [37,64,65]. However, azelastine has a bitter taste that can be bothersome to some patients (which has been corrected in newer preparations) and was mildly sedating in some clinical trials, although not in others [66]. As with glucocorticoids, we instruct patients to titrate to the lowest effective dose.

**Minimally sedating oral antihistamines** — We add minimally sedating oral antihistamines for patients who do not obtain sufficient relief with nasal glucocorticoid and antihistamine sprays ( table 6). Oral antihistamines typically reduce itching, sneezing, and rhinorrhea but are less effective for nasal congestion compared with glucocorticoid sprays.



Dosing for commonly used, minimally sedating oral antihistamines

Table 6 - larger image below

H1 antihistamines are divided into first-, second-, and third-generation agents. The second- and third-generation agents are minimally sedating and are preferred over first-generation agents because they are equally potent with fewer central nervous system effects.

- **Choice of agent, dosing, and onset of action –** Among the second-generation antihistamines, all agents appear to be similarly efficacious [1,67-71]. Rupatadine (not available in the US) has the most clinical trial data in the pediatric population [72,73]. Cetirizine appears to be the most sedating of this group; loratadine, fexofenadine, and bilastine (not available in the US) may be better options in patients who prefer to avoid any sedating side effects.

  The specific allergic sensitivities a patient has do not impact the antihistamine that is prescribed. Some agents are marketed as effective for perennial or seasonal or indoor or outdoor allergies, but marketing simply reflects the clinical trials that were performed.

  Minimally sedating antihistamines are dosed once or twice daily. Twice-daily dosing (ie, twice the recommended maximum dose) can be helpful for severe symptoms, with little risk of sedation. However, evidence is lacking regarding whether doses of second-generation antihistamines greater than those maximally recommended provide additional benefit for allergic rhinitis, and higher-than-recommended doses are associated with sedation in some cases [74].

  The onset of action of the oral second-generation agents is within one hour for most, and peak serum levels are attained in two to three hours [75,76].

  If one minimally sedating oral antihistamine at twice the recommended maximum dose is not enough to control symptoms, it is unlikely that another antihistamine agent would be more efficacious, and we switch to or add an additional treatment.

- **Efficacy –** Although second-generation antihistamines are a popular option for many patients, especially those with mild or intermittent symptoms, the combination of glucocorticoid nasal sprays and oral antihistamines has not shown clear advantage over glucocorticoid sprays alone in most studies [1,77]. Antihistamines are less effective than glucocorticoid nasal sprays, especially for the relief of nasal congestion [19,27]. Second-generation antihistamines are equally or more efficacious than cromolyn in relieving symptoms.

- **Adverse effects –** Common side effects of minimally sedating antihistamines include sedation and anticholinergic effects. Weight gain and new-onset seizures have also been reported. There is no evidence that pharmacologic tolerance develops to antihistamines [78].

  - **Sedation –** The second-generation antihistamines are minimally sedating compared with the first-generation agents. Cetirizine is sedating for approximately 10 percent of patients [79,80]. Loratadine is nonsedating for most adults at the

customary dose of 10 mg once daily, although sedation can occur at higher doses [80]. Fexofenadine is nonsedating at recommended doses and even at higher-than-recommended doses [79,81,82].

- **Anticholinergic effects –** Second-generation antihistamines have varying degrees of anticholinergic effects. Drying of the eyes, in particular, is noticeable to some patients [83].
- **Possible weight gain –** Some oral antihistamines may be associated with weight gain, although it is unclear if this is due to stimulation of appetite or reduced activity secondary to sedation and fatigue. Like other central nervous system effects, weight gain is more prominent with the older, first-generation agents, although it can occur in some patients with the second-generation agents [84]. Weight gain would be predicted to be minimal with fexofenadine [85], although we are aware of no studies directly assessing the second-generation agents for this effect. Our experience suggests that increased hunger and weight gain with nonsedating antihistamines is minimal to none in most patients.
- **New-onset seizures (observational reports) –** Very limited data suggest that therapeutic doses of second-generation antihistamines might rarely contribute to or cause new-onset seizures, as also reported with first-generation antihistamines [86].

**Combined oral antihistamine/decongestant** — Nonsedating antihistamines combined with the decongestant pseudoephedrine provide better symptom relief than either agent alone [87]. However, adverse effects of the decongestant component limit their use in some patients (table 8). Available second-generation combination agents include loratadine-pseudoephedrine, cetirizine-pseudoephedrine, and fexofenadine-pseudoephedrine.

Decongestants have a variety of adverse effects, including hypertension, insomnia, irritability, and headache (table 8). In the United States, misuse of pseudoephedrine (both as a stimulant in athletics and in the illegal production of methamphetamines) has led to the substitution of phenylephrine for pseudoephedrine in many over-the-counter first-generation combination preparations. However, the effects of phenylephrine at the 10 mg dose that is commonly available over the counter are similar to those of placebo [88-91].

Decongestants are relatively contraindicated in patients with hypertension, contraindicated in those receiving monoamine oxidase inhibitor therapy, and should be used with caution in patients with closed-angle glaucoma, cardiovascular or cerebrovascular disease, hyperthyroidism, or bladder neck obstruction.

**Therapies with limited roles** — Sedating antihistamines, combined glucocorticoid/decongestant nasal sprays, cromolyn sodium, ipratropium bromide, decongestant sprays, montelukast, and systemic glucocorticoids have limited roles in treating allergic rhinitis.

- **Sedating antihistamines –** First-generation antihistamines are all sedating and include diphenhydramine, chlorpheniramine, hydroxyzine, and brompheniramine. Although patients often use these agents initially, they should have a limited role in the treatment of most patients because of their numerous adverse effects (table 8). We strongly prefer second- and third-generation antihistamines when oral preparations are desired. They are similarly efficacious, with only minor differences [1]. Providers can offer risk-benefit counseling to caregivers as first-generation agents are widely available without a prescription.

  First-generation antihistamines cause significant sedation because they are lipophilic and easily cross the blood-brain barrier [92]. Central nervous system symptoms are reported by 20 percent or more of patients, and adverse effects on intellectual and motor function are well documented, even in the absence of subjective awareness of sedation [93-95].

  Impairments affect driving performance and have been implicated in fatal motor vehicle accidents [93,96-99]. First-generation antihistamines are prohibited in many states for some transportation workers (eg, pilots, bus drivers), and individuals taking these agents are considered to be under the influence of drugs [100]. Despite this, a report from the Federal Aviation Administration noted increasing pilot use of these drugs over the past 15 years and reported detection of these agents in 4 and 11 percent of fatalities/accidents in 1990 and 2004, respectively [101]. Other measures of cognitive effects include performance defects on tests of divided attention, working memory, vigilance, and speed [1,102,103].

  First-generation antihistamines are also problematic in children. In school-aged children, both untreated allergic rhinitis and the use of sedating antihistamines are associated with impaired school performance [96,103]. In very young children, first-generation antihistamines can cause paradoxical agitation and have been associated with new-onset seizures [86].

- **Cromolyn sodium nasal spray –** Cromolyn sodium is a mast cell stabilizer that may be a helpful option for patients who do not tolerate other agents or are concerned about side effects of other agents [104]. It has an excellent safety profile but is clearly

less effective than glucocorticoid nasal sprays or second-generation antihistamines [105]. Its utility is also limited by the need for frequent dosing (usually three to four times daily).

- **Montelukast** – Due to the potential for serious neuropsychiatric adverse effects, we typically restrict the use of montelukast for allergic rhinitis to patients with coexisting asthma who do not tolerate or prefer not to use nasal sprays and to those who do not respond adequately to oral antihistamines. We inform all patients about the potential side effects and advise them to stop the medication if they perceive adverse mood effects. (See "Antileukotriene agents in the management of asthma", section on 'Adverse effects'.)
  - **Adverse effects –** Neuropsychiatric changes have been reported in association with montelukast, including dream abnormalities, insomnia, anxiety, depression, suicidal thinking, and, in rare cases, suicide. A boxed warning was added to the product insert in 2020 with a recommendation from the US Food and Drug Administration (FDA) to avoid the use of montelukast in patients with allergic rhinitis or mild asthma in favor of other treatments.
  - **Mechanism and efficacy –** Montelukast is a leukotriene receptor antagonist that is used in the management of asthma and also has efficacy for allergic rhinitis, although the effects are modest. For the symptoms of allergic rhinitis, montelukast is similarly effective to oral antihistamines but less effective than glucocorticoid nasal sprays [27,106-110].
- **Nasal decongestant sprays** – Nasal decongestant sprays are highly effective for nasal congestion, but they are **not recommended** in the chronic treatment of allergic rhinitis, because they can cause tachyphylaxis and rhinitis medicamentosa. Topical vasoconstrictor decongestants available include phenylephrine, oxymetazoline, xylometazoline, and naphazoline. Oxymetazoline is particularly popular due to its 12-hour duration of effect.
  - **Adverse effects**:
    - **Rhinitis medicamentosa –** Rhinitis medicamentosa is a disorder arising from the downregulation of alpha-adrenergic receptors, resulting in rebound nasal congestion. It typically arises after three to seven days of more than once-daily use and can lead to a cycle of nasal congestion both caused by and temporarily relieved by the medication, resulting in escalating use and eventual dependency. (See "An overview of rhinitis", section on 'Nasal decongestant sprays'.)
    - **Other** – Some patients may experience nasal burning and dysgeusia with decongestant sprays.
  - **Potential situations in which use may be appropriate –** Nasal decongestants may be helpful when used for allergic rhinitis just before air travel in patients who have difficulties with middle ear and/or sinus equilibration with flying or in patients who have problems with altitude changes. In addition, they may be helpful for short-term use as pretreatment for other nasal therapies in patients with severe nasal obstruction. (See 'Optimal technique and use' above.)
- **Combination glucocorticoid/decongestant sprays –** Some data indicate that the combination of glucocorticoid/decongestant nasal spray may be more effective than the use of either agent alone. In one randomized trial of 60 adults with perennial allergic rhinitis, those who received once-nightly treatment with combination fluticasone and oxymetazoline reported improved symptom scores compared with those who received either agent alone or placebo [111]. There was no evidence of rhinitis medicamentosa in either decongestant nasal spray group. Further study is needed, although we have noted preliminary success in several adult patients in our clinical practice. We attempt to discontinue the decongestant spray once symptoms are controlled.
- **Systemic glucocorticoids –** Short courses (ie, a few days) of oral glucocorticoids usually abolish symptoms of allergic rhinitis and may be indicated for very severe allergic rhinitis symptoms that are preventing the patient from sleeping or working [5]. This approach was more widely used before nasal glucocorticoids and nonsedating antihistamines became available. However, systemic glucocorticoids should not be given repeatedly or for prolonged periods of time for the management of allergic rhinitis [112]. Similarly, we do **not** endorse injections of long-acting glucocorticoids for this condition, because of unpredictable absorption and the inability to dose adjust if side effects occur. (See "Major adverse effects of systemic glucocorticoids" and "Prevention and treatment of glucocorticoid-induced osteoporosis".)

## SPECIAL POPULATIONS

**Young children (<2 years of age)** — Allergic rhinitis is less prevalent in children under two years of age because the development of allergic rhinitis requires repeated exposure to inhaled allergens and takes time to develop. If a child under two years appears to have persistent nasal symptoms, other disorders should be considered, such as "daycare syndrome" (ie, repeated viral infections of the upper respiratory tract), adenoidal hypertrophy, or chronic rhinosinusitis. (See "Chronic rhinosinusitis: Clinical

manifestations, pathophysiology, and diagnosis" and "The pediatric physical examination: HEENT", section on 'Adenoidal hypertrophy'.)

If a child <2 years of age is determined to have allergic rhinitis after an evaluation for other causes, initial treatment is usually with a minimally sedating antihistamine. In the US, cetirizine and fexofenadine are approved for children age ≥6 months; these agents are available in liquid formulations, which may be helpful for young children. Sedating antihistamines (eg, diphenhydramine, chlorpheniramine) should be **avoided** in young children. These agents can cause paradoxical contraction and have been associated with an increased risk of seizures in children ≤2 years [113]. In addition, over-the-counter cold remedies containing them have been linked to a small number of deaths in children <2 years of age [114].

For children with moderate-to-severe symptoms who do not respond to minimally sedating antihistamines, changing to a glucocorticoid nasal spray is the next step ( table 5). Mometasone furoate, fluticasone furoate, and triamcinolone acetonide are approved by the US Food and Drug Administration (FDA) for use in children ≥2 years [115,116]. All choices are safe in young children with little differences between them, except some have more scent and volume than others (eg, fluticasone) and therefore may be less accepted by young children. The dose for each of these agents in young children is one spray per nostril once daily. If necessary, two sprays per nostril can be tried for a limited period (we limit to two weeks and reassess symptoms) in view of the age-limited dose approval and the concern for potential systemic side effects (eg, adrenal suppression) in children at the higher dose.

Although allergen immunotherapy is an appropriate option for most children with moderate-to-severe symptoms, it is generally not used for children younger than five years of age. (See "Subcutaneous immunotherapy (SCIT) for allergic rhinoconjunctivitis and asthma: Indications and efficacy", section on 'Age of patient' and "Sublingual immunotherapy for allergic rhinitis and conjunctivitis", section on 'Indications and patient selection'.)

**Pregnancy** — For patients who are pregnant, we first ensure they are maximizing nonpharmacologic options, including allergen avoidance and nasal saline. For those who need pharmacotherapy, second-generation antihistamines and/or cromolyn nasal spray are appropriate choices. The treatment of allergic rhinitis during pregnancy is summarized in the algorithm ( algorithm 1) and reviewed in greater detail separately. (See "Recognition and management of allergic disease during pregnancy", section on 'Allergic rhinitis/conjunctivitis'.)

**Lactation** — LactMed is an online database of information on medications and lactation hosted by the US National Library of Medicine [117]. The safety of various medications during lactation is summarized in the table ( table 9). General guidelines have been proposed [118]:

- All patients should practice allergen avoidance. (See "Allergen avoidance in the treatment of asthma and allergic rhinitis".)
- Nasal saline sprays or irrigation can always be tried for any nonspecific relief it can provide. (See 'Nasal saline' above.)
- Mild, persistent symptoms (symptoms at least four days/week and for at least four weeks/year) may be treated with intranasal budesonide or cromolyn and supplemented by cetirizine or loratadine.
- Moderate-to-severe persistent symptoms may be treated with maintenance intranasal budesonide and supplemented as needed with cetirizine or loratadine. Those with ongoing symptoms may require immunotherapy. (See 'Suboptimal management with standard therapy' below.)

**Older adults** — Glucocorticoid nasal sprays are the first-line agents for older adults with allergic rhinitis ( table 5). Epistaxis can be particularly common and bothersome for older adults, as discussed above (see 'Safety and adverse effects' above). Minimally sedating antihistamines can be used safely in some older adult patients, although a few may experience adverse effects, and slowed metabolism may warrant lower starting doses [119]. Antihistamine nasal sprays are also a good option.

We avoid first-generation sedating antihistamines in older adults as they are more susceptible to the anticholinergic effects, including dry mouth and eyes, urinary hesitancy, and confusion ( table 8) [120,121].

**Patients with comorbid conditions**

**Antihistamine eye drops for allergic conjunctivitis** — In patients with both allergic rhinitis and allergic conjunctivitis, we prefer a combination of a glucocorticoid nasal spray and ophthalmic antihistamine drops, such as olopatadine, epinastine, bepotastine, azelastine, or emedastine (where available), rather than glucocorticoid spray plus oral antihistamines ( table 10).

A small number of randomized trials have shown the addition of antihistamine eye drops to be more effective and cause less ocular drying than the addition of oral antihistamines [122,123]. However, oral antihistamines can be more practical in children

who often object to administration of eye drops.

Some glucocorticoid nasal sprays (eg, mometasone furoate and fluticasone furoate) have been shown to have a small but statistically significant effect on allergic eye symptoms [124-127]. However, many patients require an additional agent for adequate relief. The management of allergic conjunctivitis is discussed in detail separately. (See "Allergic conjunctivitis: Management", section on 'Antihistamines with mast cell-stabilizing properties'.)

**Biologics for moderate-to-severe asthma** — Up to 40 percent of patients with allergic rhinitis have concomitant asthma. Although there are no biologic therapies approved specifically for allergic rhinitis, these agents are available for moderate-to-severe asthma that is not controlled with high doses of inhaled glucocorticoids and certain biologics (eg, omalizumab) also improve the symptoms of allergic rhinitis [128,129]. Use of biologics for asthma is reviewed separately. (See "Treatment of severe asthma in adolescents and adults", section on 'Biologic agent selection'.)

## WHEN TO REFER

**Candidates for allergen immunotherapy** — Allergen immunotherapy is most beneficial for patients who are suboptimally managed with the standard therapies described above, as well as for children with moderate-to-severe or persistent symptoms. However, it can be used for any severity of allergic rhinitis, and a referral to an allergist is reasonable for any patient who is interested in this treatment option. (See "Subcutaneous immunotherapy (SCIT) for allergic rhinoconjunctivitis and asthma: Indications and efficacy" and "Sublingual immunotherapy for allergic rhinitis and conjunctivitis".)

**Suboptimal management with standard therapy** — Patients who are suboptimally managed with standard therapies are candidates for allergen immunotherapy. This includes:

- **Patients with refractory disease –** Patients are considered refractory if they have persistent symptoms despite one to two months of consistent use of a glucocorticoid spray and at least one additional agent. Such patients should first be carefully evaluated for alternative etiologies, such as chronic rhinosinusitis, rhinitis due to medications, and rhinitis due to systemic diseases, prior to referral for immunotherapy. (See "Allergic rhinitis: Clinical manifestations, epidemiology, pathogenesis, and diagnosis", section on 'Differential diagnosis'.)

- **Patients with complicated disease –** This includes patients with recurrent episodes of sinusitis or otitis media and those who have required systemic glucocorticoids to control symptoms. In some cases (eg, recurrent sinusitis), referral to an otolaryngologist to evaluate for structural etiologies may be helpful prior to consideration of allergen immunotherapy.

- **Barriers to standard therapies –** This includes patients who experience bothersome adverse effects from standard therapies or who have other barriers to use, such as cost or medication burden.

**Children with moderate-to-severe or persistent symptoms** — Children with moderate-to-severe or persistent allergic rhinitis (table 3) benefit from allergen immunotherapy because it has been shown to help alter the progression of allergic disease and decrease the risk of developing asthma. (See "Subcutaneous immunotherapy (SCIT) for allergic rhinoconjunctivitis and asthma: Indications and efficacy", section on 'Prevention of asthma' and "Sublingual immunotherapy for allergic rhinitis and conjunctivitis", section on 'Reduced risk of asthma development'.)

In general, both subcutaneous and sublingual immunotherapy are only used for children ≥5 years old as younger children may have difficulty tolerating the treatments and/or communicating their symptoms. (See "Subcutaneous immunotherapy (SCIT) for allergic rhinoconjunctivitis and asthma: Indications and efficacy", section on 'Age of patient'.)

**Consideration of allergy testing** — Patients whose management might be enhanced by identification of allergic triggers can benefit from referral to an allergist for consideration of allergy testing. This includes patients who are motivated to adjust their environment based on allergy testing or are unclear about what their triggers are and would like more information.

Patients who are able to discontinue antihistamines without experiencing intolerable symptoms should do so at least one week prior to their appointment in case skin testing is indicated at the initial visit. Intranasal glucocorticoids and asthma medications do not interfere with skin testing and should **not** be discontinued. Patients with questions concerning which medications to continue or withhold should be instructed to contact the specialist's office prior to their appointment. (See "Overview of skin testing for IgE-mediated allergic disease".)

## SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "Society guideline links: Rhinitis".)

## INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or email these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient education" and the keyword(s) of interest.)

- Basics topics (see "Patient education: Environmental allergies in adults (The Basics)" and "Patient education: Giving your child medicines (The Basics)" and "Patient education: Environmental allergies in children (The Basics)" and "Patient education: How to use nasal medicines (The Basics)")
- Beyond the Basics topics (see "Patient education: Allergic rhinitis (Beyond the Basics)" and "Patient education: Trigger avoidance in allergic rhinitis (Beyond the Basics)")

## SUMMARY AND RECOMMENDATIONS

- **Nonpharmacologic measures for all patients –** Nonpharmacologic strategies are helpful for any severity level and are advised regardless of the pharmacotherapy used.
  - We advise allergen avoidance when feasible (table 1). (See 'Allergen avoidance' above and "Allergen avoidance in the treatment of asthma and allergic rhinitis".)
  - We suggest nasal saline spray or irrigation (**Grade 2C**), either alone or in combination with pharmacotherapy. Irrigation instructions are provided in the table (table 2). (See 'Nasal saline' above.)
- **Treatment options –** Glucocorticoid nasal sprays are the most effective single-agent therapy for allergic rhinitis. Glucocorticoid nasal sprays are also available as combination sprays with nasal antihistamines, although in some countries (including the United States) these require prescriptions. Several glucocorticoid nasal sprays, such as fluticasone propionate, mometasone furoate, and fluticasone furoate, have minimal systemic bioavailability and are conveniently dosed once daily. A summary of available agents is shown in the table (table 5). We start at the maximal recommended dose for age and then taper to the lowest effective dose once symptoms are controlled.

  Although oral antihistamines are frequently used, they are not as effective as glucocorticoid nasal spray (table 4). When they are used, we prefer minimally sedating (second-generation) oral antihistamines, as they are equally efficacious as first-generation but with fewer side effects (table 6).
- **Mild, intermittent, or seasonal symptoms (≥2 years of age) –** For adults and children ≥2 years of age with mild, intermittent, or seasonal symptoms, we suggest a glucocorticoid nasal spray (**Grade 2B**). This can be combined with a nasal antihistamine (eg, azelastine or olopatadine; for patients ≥6 years) if available, and used regularly or only as needed. For predictable exposures, we start therapy two days before and continue for two days after the exposure. For seasonal symptoms, we initiate regular treatment at least one week prior to pollen season.

  Alternatives include regular or as-needed antihistamine-only sprays, minimally sedating oral antihistamines (at least two hours before exposure), or cromolyn nasal spray (at least 30 minutes before exposure). (See 'Mild or intermittent symptoms' above and 'Seasonal symptoms' above.)
- **Persistent or moderate-to-severe symptoms (≥2 years of age) –** For adults and children ≥2 years of age with moderate-to-severe symptoms (table 3), we recommend a glucocorticoid nasal spray (**Grade 1A**), which can be used with a nasal antihistamine (eg, azelastine or olopatadine; for patients ≥6 years) if available. We start at the maximal recommended dose for age and then taper to the lowest effective dose once symptoms are controlled.

Most patients experience significant relief of symptoms; glucocorticoid sprays are particularly effective for nasal congestion and postnasal drainage. For patients who do not respond sufficiently to glucocorticoid-only sprays, we suggest adding an antihistamine nasal spray rather than other agents (**Grade 2B**). Additional alternatives include minimally-sedating oral antihistamines or combined oral antihistamine/decongestant agents. (See 'Persistent or moderate-to-severe symptoms' above.)

- **Special populations**:
  - **Children <2 years of age –** Allergic rhinitis is uncommon in children <2 years of age, and patients should be evaluated for other disorders, including "daycare syndrome" (ie, repeated viral upper respiratory infections), adenoidal hypertrophy, and chronic rhinosinusitis. We have a low threshold for referring young children with apparent allergic rhinitis symptoms to otolaryngology for further evaluation. (See 'Young children (<2 years of age)' above.)
  - **Pregnancy and lactation –** During pregnancy and lactation, we maximize nonpharmacologic therapy and use minimally sedating oral antihistamines or minimally-bioavailable glucocorticoid nasal sprays (eg, fluticasone, mometasone) ( algorithm 1). For those who are concerned about glucocorticoids, cromolyn nasal spray is another safe option, although it is less effective. (See "Recognition and management of allergic disease during pregnancy" and 'Lactation' above.)
  - **Older adults –** Glucocorticoid nasal sprays are the first-line agents for older adults with allergic rhinitis. We avoid first-generation antihistamines in this group because of cholinergic side effects (eg, dry mouth, urinary retention, confusion). Minimally sedating antihistamines can be used in some older adults, although dose reduction may be necessary for those who experience adverse effects. (See 'Older adults' above.)
  - **Comorbid allergic conjunctivitis –** For patients with both allergic rhinitis and allergic conjunctivitis whose symptoms are not controlled with a glucocorticoid nasal spray, we add antihistamine eye drops ( table 10). (See "Allergic conjunctivitis: Management", section on 'Topical therapies'.)
  - **Comorbid asthma –** For those with comorbid moderate-to-severe asthma, treatment (eg, omalizumab) can also improve allergic rhinitis symptoms. (See 'Biologics for moderate-to-severe asthma' above.)
- **Indications for specialty referral –** We refer the following patients to an allergist:
  - Patients who are candidates for allergen immunotherapy. This includes:
    - Those with persistent symptoms despite one or two months of consistent use of a glucocorticoid spray and at least one additional agent, or with recurrent sinusitis or otitis media.
    - Children with moderate-to-severe or persistent allergic rhinitis, in whom allergen immunotherapy may help prevent progression to asthma. (See 'Candidates for allergen immunotherapy' above.)
    - Those who have barriers to standard therapies or are interested in a nondrug approach.
  - Patients who would benefit from allergy testing, such as those who have unclear triggers and are motivated to adjust their environment based on results. (See 'Consideration of allergy testing' above.)

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Topic 7526 Version 68.0

## GRAPHICS

Table 1: **Basic measures to control exposure to allergens**

| Indoor allergen | Recommendations for reducing exposure |
|---|---|
| Animal dander | ▪ Remove animal from house, or at minimum, keep animal out of patient's bedroom. Keep pet in a room with a HEPA filter, and replace the filter as recommended by the manufacturer.<br>▪ Cover air ducts that lead to bedroom with filters. Replace filters as recommended by the manufacturer.<br>▪ Use air filters and vacuums with HEPA filters. Replace the filter as recommended by the manufacturer. |
| Dust mites | ▪ Less costly:<br>• Encase mattress, pillows, and box spring in allergen-impermeable covers. Finely woven covers for pillows and duvets are preferable.<br>• Wash bedding weekly in warm water with detergent or use electric dryer on hot setting.<br>• Reduce indoor humidity to <50%. Use air conditioners in warm climates. Avoid living in basements because controlling humidity is very difficult.<br>▪ More costly:<br>• Remove fitted carpets from the bedroom. If desired, use area rugs that can be picked up and cleaned.<br>• Replace old upholstered furniture with leather, vinyl, or wood. |
| Cockroaches | ▪ Use poison bait or traps to control. Consult professional exterminator for severe infestation.<br>▪ Periodically clean home thoroughly.<br>▪ Encase all food fully, and do not store garbage or papers inside the home.<br>▪ Fix water leaks. |
| Indoor mold | ▪ Clean moldy surfaces with dilute bleach solution.<br>▪ Fix water leaks.<br>▪ Reduce indoor humidity to <50%. Avoid use of humidifiers. Use air conditioners in warm climates. Avoid living in basements.<br>▪ Evaporative (or swamp) coolers should be avoided or cleaned regularly. |
| Rodents | ▪ Consult a professional exterminator.<br>▪ Periodically clean home thoroughly.<br>▪ All food should be stored in sealed containers. Do not store garbage inside.<br>▪ Repair holes in walls, doors, floors, and block other entry points. |
| Outdoor allergens | ▪ Limit time outdoors during the pollen seasons, especially in the morning when pollen counts are highest.<br>▪ Keep car windows closed, and use the air conditioner to filter the air.<br>▪ Keep windows of buildings closed, and use the air conditioner or room purifiers with HEPA filters (avoid "ionic" air filters, which produced the respiratory irritant ozone).<br>▪ Wear a high-quality mask if spending prolonged time outdoors.<br>▪ Wear glasses or sunglasses to minimize pollen in the eyes.<br>▪ Avoid hanging clothes outside to dry (pollen may adhere to fabric).<br>▪ After being outdoors, shower to remove pollen from hair and skin, and change clothes. |

The measures that can effectively reduce exposure are different for each allergen. Those listed are suggested but have not been verified to be effective in isolation. Studies showing greatest benefit from allergen remediation were those that adapted the measures to each patient's individual sensitivities and addressed all of a patient's allergies in parallel.

HEPA: high-efficiency particulate air.

*Reference:*

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*Adapted from: American Academy of Allergy, Asthma, & Immunology. The Allergy Report, American Academy of Allergy, Asthma, & Immunology 2000.*

Graphic 58870 Version 7.0

Table 2: **How to perform nasal irrigation**

| **Buffered normal saline nasal irrigation** |
| --- |
| **The benefits** |
| ▪ Saline (saltwater) washes the mucus and irritants from your nose.<br>▪ The sinus passages are moisturized.<br>▪ Studies have also shown that a nasal irrigation improves the function of cilia (tiny hair-like structures on cells that move the mucus). |
| **The recipe** |
| 1. Use a one-quart glass jar that is thoroughly cleansed.<br>2. You may use a large medical syringe (30 cc), water pick with an irrigation tip, squeeze bottle, or Neti pot. Do not use a baby bulb syringe. The syringe or pick should be sterilized frequently or replaced every 2 to 3 weeks to avoid contamination and infection.<br>3. Fill with water that has been distilled, previously boiled, or otherwise sterilized. Plain tap water is not recommended, because it is not necessarily sterile.<br>4. Add 1 to 1½ heaping teaspoons of pickling/canning salt. Do **not** use table salt, because it contains a large number of additives.<br>5. Add 1 teaspoon of baking soda (pure bicarbonate).<br>6. Mix ingredients together and store at room temperature. Discard after 1 week.<br>7. You may also make up a solution from premixed packets that are commercially prepared specifically for nasal irrigation. |
| **The instructions** |
| Irrigate your nose with saline 1 to 2 times per day.<br>▪ If you have been told to use nasal medication, you should always use your saline solution first. The nasal medication is much more effective when sprayed onto clean nasal membranes, and the spray will reach deeper into the nose.<br>▪ Pour the amount of fluid you plan to use into a clean bowl. Do **not** put your used syringe back into the storage container, because it contaminates your solution.<br>▪ You may warm the solution slightly in the microwave but be sure that the solution is **not hot**.<br>▪ Bend over the sink (some people do this in the shower) and squirt the solution into each side of your nose, aiming the stream toward the back of your head, **not** the top of your head. The solution should flow into one nostril and out of the other, but it will not harm you if you swallow a little. Once all the solution has run out of your nose, blow your nose gently. However, if fluid gets trapped in your ears, avoid blowing your nose for about 15 minutes.<br>▪ Some people experience a little burning sensation the first few times that they use buffered saline solution, but this usually goes away after they adapt to it. |

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Graphic 71059 Version 18.0

Table 3: **Classification of allergic rhinitis**

| |
|---|
| "Intermittent" means that the symptoms are present:<br>▪ Less than 4 days a week<br>▪ Or for less than 4 weeks |
| "Persistent" means that the symptoms are present:<br>▪ More than 4 days a week<br>▪ And for more than 4 weeks |
| "Mild" means that none of the following items are present:<br>▪ Sleep disturbance<br>▪ Impairment of daily activities, leisure, and/or sport<br>▪ Impairment of school or work<br>▪ Troublesome symptoms |
| "Moderate-severe" means that one or more of the following items are present:<br>▪ Sleep disturbance<br>▪ Impairment of daily activities, leisure, and/or sport<br>▪ Impairment of school or work<br>▪ Troublesome symptoms |

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Table 4: **Choice of medications for allergic rhinitis**

| Medication | Commonly used examples | Pros | Cons |
|---|---|---|---|
| **Glucocorticoid spray** | Fluticasone propionate (brand name Flonase)<br>Mometasone (brand name Nasonex) | More effective for congestion and nasal drainage than antihistamines<br>Low systemic bioavailability | Onset of action several hours |
| **Combination glucocorticoid/antihistamine nasal spray** | Azelastine-fluticasone (brand name Dymista)<br>Olopatadine-mometasone (brand name Ryaltris) | More efficacious than either agent alone<br>Rapid onset due to antihistamine component | Cannot be used for young children<br>Requires a prescription (in the United States)<br>Bitter taste |
| **Antihistamine-only nasal spray** | Azelastine (brand name Astepro)<br>Olopatadine (brand name Patanase) | Rapid onset of action (15 minutes) | Bitter taste |
| **Minimally sedating oral antihistamines** | Cetirizine (brand name Zyrtec)<br>Bilastine (not available in the United States) | Effective for itching, sneezing, and watery rhinorrhea | Less effective for nasal congestion compared with glucocorticoid sprays |
| **Combined oral antihistamine/decongestant** | Loratadine-pseudoephedrine (example brand name Claritin-D)<br>Cetirizine-pseudoephedrine (example brand name Zyrtec-D) | More efficacious than either agent alone | Decongestant component has significant adverse effects including hypertension, insomnia, irritability, headache |

This table reviews the preferred agents for allergic rhinitis with common examples, benefits, and drawbacks.

Graphic 148386 Version 1.0

Table 5: **Glucocorticoid nasal sprays for treatment of rhinitis**

| Drug | Common brand name(s) and strength | Generic available | Available without a prescription (OTC) | Usual adult dose per nostril | Lower age limit based on FDA labeling | Usual pediatric dose per nostril | Type of preparation (alcohol content)* | Notes |
|---|---|---|---|---|---|---|---|---|
| **Single-ingredient** | | | | | | | | |
| Beclomethasone | Pediatric: Qnasl Children's (40 mcg/spray)<br>Adolescent/adult: Qnasl (80 mcg/spray) | No | No | 2 sprays once daily using 80 mcg/spray product | 4 years | 4 to 11 years: 1 spray once daily using 40 mcg/spray product<br>≥12 years: 2 sprays once daily using 80 mcg/spray product | Pressurized aerosol spray (8% alcohol*) | |
| | Beconase AQ (50 mcg/spray)<br>Available in some areas (not United States or Canada) | No | No | 1 to 2 sprays twice daily | 6 years | 1 to 2 sprays twice daily<br>In children 6 to 11 years, start with 1 spray twice daily | Aqueous suspension pump spray (0.25% alcohol) | |
| Budesonide¶ | Generic (formerly Rhinocort Allergy) (OTC) (32 mcg/spray) | Yes | Yes | 1 to 2 sprays once daily | 6 years | 1 to 2 sprays once daily<br>In children 6 to 11 years, start with 1 spray once daily | Aqueous suspension pump spray | |
| Ciclesonide | Omnaris (50 mcg/spray) | No | No | 2 sprays once daily | 6 years[Δ] | 6 to 11 years: 1 or 2 sprays once daily<br>≥12 years: 2 sprays once daily | Aqueous suspension pump spray | Preferred option in children due to minimal systemic bioavailability |
| Flunisolide | Generic (formerly Nasalide) (25 mcg/spray) | Yes | No | 2 sprays 2 or 3 times daily (maximum 2 sprays 4 times daily) | 6 years | 6 to 14 years: 1 spray 3 times daily or 2 sprays twice daily<br>≥15 years: 2 sprays 2 or 3 times daily (maximum 2 sprays 4 times daily) | Aqueous suspension pump spray (contains propylene glycol, a possible irritant) | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Fluticasone furoate[¶] | Flonase Sensimist (OTC) (27.5 mcg/spray) | No | Yes | 2 sprays once daily | 2 years | 1 or 2 sprays once daily<br>In children 2 to 11 years, start with 1 spray once daily | Aqueous suspension pump spray | Preferred option in children due to minimal systemic bioavailability |
| Fluticasone propionate[¶] | Flonase Allergy Relief (OTC) (50 mcg/spray) | Yes | Yes | 2 sprays once daily or 1 spray twice daily | 4 years | 4 to 11 years: 1 spray once daily<br>≥12 years: 2 sprays once daily or 1 spray twice daily | Aqueous suspension pump spray (0.25% alcohol) | Preferred option in children due to minimal systemic bioavailability |
| Mometasone[¶] | Nasonex 24HR Allergy (OTC) (50 mcg/spray) | Yes | Yes | 2 sprays once daily | 2 years | 2 to 11 years: 1 spray once daily<br>≥12 years: 2 sprays once daily | Aqueous suspension pump spray | Preferred option in children due to minimal systemic bioavailability |
| Triamcinolone[¶] | Nasacort Allergy 24HR (OTC) (55 mcg/spray) | Yes | Yes | 2 sprays once daily | 2 years | 2 to 5 years: 1 spray once daily<br>≥6 years: 1 to 2 sprays once daily<br>In children 6 to 11 years, start with 1 spray once daily | Aqueous suspension pump spray | Avoid in pregnancy due to preliminary reports of fetal respiratory malformations |
| **Combination with antihistamine (faster acting than glucocorticoid alone; may help patients who don't respond fully to single-ingredient glucocorticoid sprays)** | | | | | | | | |
| Azelastine-fluticasone propionate[¶] | Dymista (azelastine 137 mcg/fluticasone propionate 50 mcg per spray) | Yes | No | 1 spray twice daily | 6 years[Δ] | ≥6 years: 1 spray twice daily | Aqueous suspension pump spray (0.25% alcohol) | |
| Olopatadine-mometasone[¶] | Ryaltris (olopatadine 665 mcg/mometasone 25 mcg per spray) | No | No | 2 sprays twice daily | 12 years[Δ] | ≥12 years: 2 sprays twice daily | Aqueous suspension pump spray | |

Nasal sprays work best when they are administered properly, and the medication remains in the nose rather than draining down the back of the throat. Note that the recommended techniques for the aqueous and aerosol sprays are different. If the nose is crusted or contains mucus, it should first be cleaned with a saline nasal spray prior to use of intranasal sprays. Some people find that holding the other nostril closed with a finger improves their ability to draw the spray into the upper nose.

Once symptoms are controlled, the daily dose can be reduced to the lowest dose that maintains control.

Dosing and product descriptions are based upon products available in the United States and some other countries. Product descriptions in other countries may differ in some detail. Consult the drug monographs included within UpToDate and local product information for additional detail.

For agent selection, refer to UpToDate content on rhinitis.

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CYP: cytochrome P450; FDA: US Food and Drug Administration; OTC: over the counter (available without a prescription in the United States and some other countries).

* Alcohol may contribute to dryness or irritation. Patients reporting these symptoms may prefer alcohol-free sprays.

¶ Subject to drug interactions with CYP3A4 inhibitors, requiring caution or avoidance. For example, fluticasone nasal sprays should be avoided in patients who are receiving strong CYP3A4 inhibitors. For specific interactions, refer to the drug interactions program included within UpToDate.

Δ Some available data suggest use may be safe in younger age groups. Refer to drug monographs for additional details.

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*Data from: UpToDate Lexidrug. More information available at https://online.lexi.com/.*

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Graphic 55833 Version 49.0

Table 6: **Dosing for commonly used, minimally sedating oral antihistamines**

| Medication | Adult dosing | Pediatric dosing |
|---|---|---|
| Cetirizine*¶ | **Standard:** 10 mg once daily<br>**New-onset urticaria:** May increase to 10 mg twice daily<br>**Chronic urticaria:** May gradually increase to 20 mg twice daily | **≥12 years:** 10 mg once daily<br>**≥6 to 11 years:** 5 to 10 mg once daily<br>**2 to <6 years:** 2.5 to 5 mg once daily or 2.5 mg twice daily<br>**6 months to 2 years:** 2.5 mg once daily<br>NOTE: For chronic urticaria and perennial rhinitis, dose may be increased to 2.5 mg twice daily for children aged 12 months to 2 years |
| Levocetirizine* | **Standard:** 5 mg once daily<br>**New-onset urticaria:** May increase to 5 mg twice daily<br>**Chronic urticaria:** May gradually increase to 10 mg twice daily | **≥12 years:** 2.5 to 5 mg once daily<br>**6 to 11 years:** 2.5 mg once daily<br>**6 months to <6 years:** 1.25 mg once daily |
| Loratadine¶ | **Standard:** 10 mg once daily<br>**New-onset urticaria:** May increase to 10 mg twice daily<br>**Chronic urticaria:** May gradually increase to 20 mg twice daily | **≥6 years:** 10 mg once daily or in 2 divided doses<br>**2 to <6 years:** 5 mg once daily |
| Desloratadine¶ | **Standard:** 5 mg once daily<br>**New-onset urticaria:** May increase to 5 mg twice daily<br>**Chronic urticaria:** May gradually increase to 10 mg twice daily | **≥12 years:** 5 mg once daily<br>**6 to 11 years:** 2.5 mg once daily<br>**1 to <6 years:** 1.25 mg once daily (requires oral solution; not commercially available in US)<br>**6 months to 1 year:** 1 mg once daily (requires oral solution; not commercially available in US) |
| Fexofenadine¶ | **Standard:**<br>▪ 24-hour formulation: 180 mg once daily<br>▪ 12-hour formulation: 60 mg twice daily<br>**New-onset or chronic urticaria:**<br>▪ 24-hour formulation: 180 mg once daily (higher doses are typically not more effective) | **≥12 years:**<br>▪ 24-hour formulation: 180 mg once daily<br>▪ 12-hour formulation: 60 mg twice daily<br>**2 to 11 years:** 30 mg twice daily<br>**6 months to 2 years and ≥10.5 kg:** 30 mg twice daily<br>**6 months to 2 years and <10.5 kg:** 15 mg twice daily |
| Ebastine∆ | **Standard:** 10 mg once daily<br>**New-onset or chronic urticaria:** May increase to 20 mg once daily | **≥12 years:** 10 mg once daily |
| Bilastine∆ | **Standard:** 20 mg once daily<br>**New-onset or chronic urticaria:** May gradually increase to 80 mg once daily | **≥12 years:** 20 mg once daily<br>**≥4 to 11 years and ≥16 kg:** 10 mg once daily |
| Rupatadine∆ | **Standard:** 10 mg once daily<br>**New-onset or chronic urticaria:** May gradually increase to 40 mg once daily | **≥12 years:** 10 mg once daily<br>**≥2 to 11 years and >25 kg:** 5 mg once daily<br>**≥2 to 11 years and 10 to ≤25 kg:** 2.5 mg once daily |

This table is to be used with UpToDate content on the treatment of allergic rhinitis, new-onset urticaria (hives), chronic spontaneous urticaria, and other allergic disorders. Standard dosing is used for treatment of allergic rhinitis and as initial dosing for urticaria. For chronic conditions, regular (daily) use is generally recommended; however, as-needed use may be sufficient for mild symptoms. Doses listed are for patients with normal kidney and liver function. Doses may need to be modified in kidney or hepatic impairment; refer to UpToDate Lexidrug monographs for details. Significant drug interactions may exist requiring dose adjustment or avoidance of some combinations; refer to the drug interactions program included within UpToDate to determine specific interactions.

---

IV: intravenous.

* Cetirizine and levocetirizine may be more sedating than others listed; they can be taken in the evening if sedating. Cetirizine is also available as an IV solution that may be useful in a health care setting when a rapid onset is needed or oral administration is not available.

¶ Also available as a single-pill combination with pseudoephedrine.

∆ Widely available in areas other than United States.

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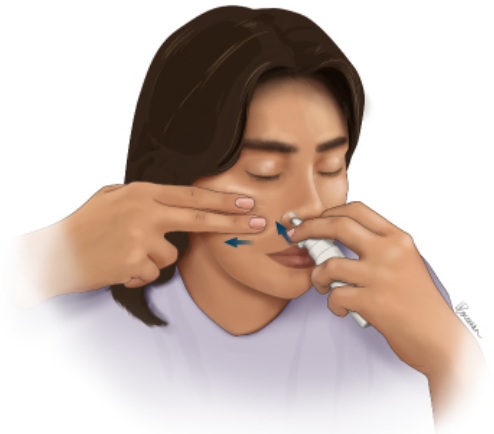
Graphic 148024 Version 5.0

Figure 1: **How to use a nasal spray properly**

1. Blow your nose gently.
2. Shake the spray bottle to make sure the medicine is well mixed. Remove the cap.
3. Spray once or twice into the air until a good mist is dispensed.
4. Tuck your chin in and tip your head forward slightly.
5. To spray in the left side, use 1 or 2 of your left fingers to widen your left nostril slightly by pulling the skin near the nostril to the side. Use your right hand to hold the nasal spray. Put the nozzle into the left nostril, and spray away from the membrane in the middle of the nose (called the nasal septum). Spraying at the septum can cause irritation and sometimes minor bleeding. Think of aiming the spray towards the outside corner of your left eye. If this hand positioning feels awkward, you can skip the step of pulling the nostril open and just spray with either hand, but it is important not to spray directly at the septum.



6. Do not sniff until the medicine is dispensed, but you can close your mouth and inhale gently to keep the medicine from running out of the nose. If you sniff too vigorously, the medicine will go down your throat, where it does nothing for your nasal symptoms.
7. To spray the right side, use 1 or 2 of your right fingers to pull your right nostril slightly open and use your left hand to hold the nasal spray. Follow the same instructions as above, pointing the spray towards your right ear.

If you are also doing saline nasal irrigations, do those first and then wait 5 to 10 minutes before using your nasal spray medication. Waiting allows the irrigation to drain completely from the nose and increases the chance that medicine sprayed into the nose afterwards will be absorbed as fully as possible by the clean nasal lining.

Graphic 141563 Version 2.0

Table 7: **Cytochrome P450 3A (including 3A4) inhibitors and inducers**

| Strong inhibitors | Moderate inhibitors | Strong inducers | Moderate inducers |
|---|---|---|---|
| ▪ Adagrasib<br>▪ Atazanavir<br>▪ Ceritinib<br>▪ Clarithromycin<br>▪ Cobicistat and cobicistat-containing coformulations<br>▪ Darunavir<br>▪ Idelalisib<br>▪ Itraconazole<br>▪ Ketoconazole<br>▪ Levoketoconazole<br>▪ Lonafarnib<br>▪ Lopinavir<br>▪ Mifepristone*<br>▪ Nefazodone<br>▪ Nelfinavir<br>▪ Nirmatrelvir-ritonavir<br>▪ Ombitasvir-paritaprevir-ritonavir<br>▪ Ombitasvir-paritaprevir-ritonavir plus dasabuvir<br>▪ Posaconazole<br>▪ Relacorilant<br>▪ Ritonavir and ritonavir-containing coformulations<br>▪ Tucatinib<br>▪ Voriconazole | ▪ Amiodarone¶<br>▪ Aprepitant<br>▪ Avacopan<br>▪ Berotralstat<br>▪ Cimetidine¶<br>▪ Conivaptan<br>▪ Crizotinib<br>▪ Cyclosporine¶<br>▪ Diltiazem<br>▪ Duvelisib<br>▪ Dronedarone<br>▪ Erythromycin<br>▪ Fedratinib<br>▪ Fluconazole<br>▪ Fosamprenavir<br>▪ Fosaprepitant¶<br>▪ Fosnetupitant-palonosetron<br>▪ Grapefruit juice<br>▪ Imatinib<br>▪ Isavuconazole (isavuconazonium sulfate)<br>▪ Lefamulin<br>▪ Letermovir<br>▪ Netupitant<br>▪ Nilotinib<br>▪ Nirogecestat<br>▪ Ribociclib<br>▪ Rilzabrutinib<br>▪ Schisandra<br>▪ Verapamil | ▪ Apalutamide<br>▪ Carbamazepine<br>▪ Encorafenib<br>▪ Enzalutamide<br>▪ Fosphenytoin<br>▪ Lumacaftor<br>▪ Lumacaftor-ivacaftor<br>▪ Mitotane<br>▪ Phenytoin<br>▪ Rifampin (rifampicin) | ▪ Bexarotene<br>▪ Bosentan<br>▪ Cenobamate<br>▪ Dabrafenib<br>▪ Dexamethasone∆<br>▪ Dipyrone<br>▪ Efavirenz<br>▪ Elagolix, estradiol, and norethindrone therapy pack◊<br>▪ Eslicarbazepine<br>▪ Etravirine<br>▪ Fexinidazole<br>▪ Lorlatinib<br>▪ Mitapivat<br>▪ Modafinil<br>▪ Nafcillin<br>▪ Pacritinib<br>▪ Pexidartinib<br>▪ Phenobarbital<br>▪ Primidone<br>▪ Repotrectinib<br>▪ Rifabutin<br>▪ Rifapentine<br>▪ Sotorasib<br>▪ St. John's wort |

- For drug interaction purposes, the inhibitors and inducers of CYP3A metabolism listed above can alter serum concentrations of drugs that are dependent upon the CYP3A subfamily of liver enzymes, including CYP3A4, for elimination or activation.
- These classifications are based upon FDA guidance.[1,2] Other sources may use a different classification system, resulting in some agents being classified differently.
- Data are for systemic drug forms. Degree of inhibition or induction may be altered by dose, method, and timing of administration.
- Weak inhibitors and inducers are not listed in this table with exception of a few examples. Clinically significant interactions can occasionally occur due to weak inhibitors and inducers (eg, target drug is highly dependent on CYP3A4 metabolism and has a narrow therapeutic index). Accordingly, specific interactions should be checked using a drug interaction program such as the drug interactions program included within UpToDate.
- Refer to UpToDate topics on specific agents and indications for further details.

CYP: cytochrome P450; FDA: US Food and Drug Administration.

* Mifepristone is a strong inhibitor of CYP3A4 when used chronically (eg, for hyperglycemia in patients with Cushing syndrome). The CYP3A4 inhibitory effect of a single 200 mg mifepristone dose is likely to be weaker and transient; however, specific data are lacking.

¶ Classified as a weak inhibitor of CYP3A4, according to FDA system.[1]

∆ Classified as a weak inducer of CYP3A4, according to FDA system.[1]

◊ The fixed-dose combination therapy pack taken in the approved regimen has moderate CYP3A4 induction effects. When elagolix is used as a single agent, it is a weak CYP3A4 inducer. Norethindrone and estradiol are not CYP3A4 inducers.

*Data from: UpToDate Lexidrug. More information available at https://online.lexi.com/.*

*References:*

1. *M12 drug interaction studies: Guidance for industry. US Department of Health and Human Services, Food and Drug Administration, Center for Drug Evaluation and Research (CDER), Center for Biologics Evaluation and Research (CBER). https://www.fda.gov/media/161199/download (Accessed on June 12, 2025). Summary available at https://www.fda.gov/media/135586/download (Accessed on June 12, 2025).*
2. *US Food and Drug Administration. Drug development and drug interactions: Table of substrates, inhibitors and inducers. https://www.fda.gov/drugs/drug-interactions-labeling/drug-development-and-drug-interactions-table-substrates-inhibitors-and-inducers (Accessed on June 12, 2025).*

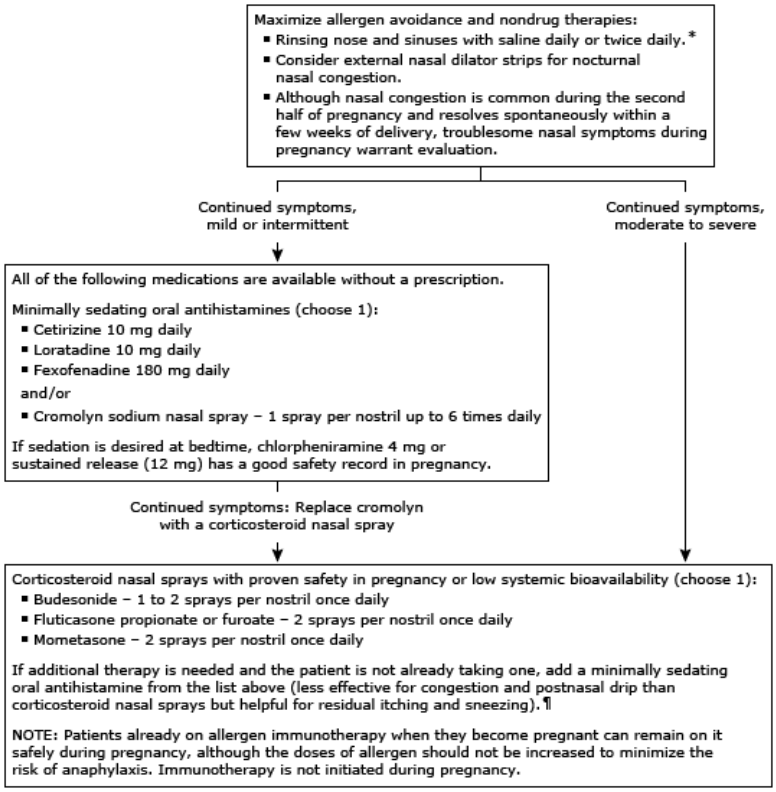
Graphic 76992 Version 112.0

Table 8: **Some adverse effects of antihistamines and decongestants**

| |
|---|
| **Antihistamines** |
| **Anticholinergic effects** |
| Dry mouth and eyes |
| Impotence |
| Urinary hesitancy |
| Glaucoma |
| **Central nervous system effects** |
| Sedation |
| Rarely stimulation (usually children) |
| Confusion (older patients) |
| Cognitive impairment |
| **Miscellaneous effects** |
| Weight gain |
| Hypersensitivity |
| Prolonged QT interval |
| Ventricular arrhythmias |
| **Decongestants** |
| Nervousness |
| Irritability |
| Insomnia |
| Headache |
| Urinary hesitancy |
| Tachycardia/palpitations |
| Hypertension |
| Nausea |

Graphic 65742 Version 4.0

Algorithm 1: **Treatment of allergic rhinitis in pregnancy**




* The nasal passages can be rinsed with a saline spray or the lower sinuses can be rinsed with larger volumes using a neti pot or squeeze bottle. Refer to UpToDate content on pharmacotherapy for allergic rhinitis, discussion of nasal saline.

¶ Medications that are generally avoided in pregnancy include oral decongestants (especially in the first trimester), antihistamine nasal sprays (due to lack of human data), and herbal therapies (not regulated and thus could contain contaminants).

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Graphic 134600 Version 3.0

Table 9: **Therapies for allergic rhinitis in lactating women**

| Drug category | Amount passed to infant in breast milk | Conclusions about maternal use during breastfeeding |
|---|---|---|
| Intranasal glucocorticoid sprays (budesonide preferred) | Minimal | Compatible. |
| Second-generation oral antihistamines | | |
| Cetirizine | Minimal to small amounts | Large doses or prolonged use may cause sedation in nursing infant. May decrease milk supply if combined with a sympathomimetic. |
| Loratadine | Minimal to small amounts | Compatible but might reduce milk supply if combined with a sympathomimetic, such as pseudoephedrine. |
| Antihistamine nasal sprays | Minimal | Probably compatible. Use intranasal glucocorticoid preferentially. |
| Oral decongestants | | |
| Pseudoephedrine | Passes into milk | Use short-acting preparations only and take just after breastfeeding to minimize amount entering milk. Use intranasal glucocorticoid preferentially for persistent congestion. Use topical vasoconstrictor nasal spray for therapy of less than four days. |
| Phenylephrine | Little information available | Not recommended. |
| Cromolyn nasal spray | Insignificant | Compatible. |
| Leukotriene-receptor antagonists | | |
| Montelukast | Unknown | Use only if other compatible agents are not sufficient and symptoms are significant. Avoid in newborn or preterm infant. |
| Zafirlukast | About 20% | Use only if other compatible agents are not sufficient and symptoms are significant. Avoid in newborn or preterm infant. |
| Ipratropium nasal spray | Unknown; maternal serum levels are negligible and milk levels will be very low, if at all; any drug in milk would not be absorbed by the infant | Probably compatible. |

*References:*

1. *Incaudo AF, Takach P. The diagnosis and treatment of allergic rhinitis during pregnancy and lactation. Immunol Allergy Clin N Am 2006; 26:137.*
2. *Briggs GG, Freeman RK, Yaffe SJ. Drugs in pregnancy and lactation, 10th ed, Lippincott Williams & Wilkins, Philadelphia 2015.*
3. *LactMed: United States National Library of Medicine. https://www.ncbi.nlm.nih.gov/books/NBK501922/.*

Graphic 58772 Version 7.0

Table 10: **Ophthalmic medications for the treatment of allergic conjunctivitis**

| Pharmacologic class | Usual adult dosing | Pediatric dosing |
|---|---|---|
| **Antihistamines with mast cell-stabilizing properties: Decreased itching should be evident within 24 to 72 hours; may result in dry eye sensation or burning** | | |
| Olopatadine 0.1% and 0.2% (OTC Pataday, generics), 0.7% (OTC Pataday) | One drop per eye twice daily (0.1%); one drop per eye once daily (0.2% and 0.7%) | ≥2 years: One drop per eye twice daily (0.1%); one drop per eye once daily (0.2% and 0.7%) |
| Alcaftadine 0.25% (OTC Lastacaft) | One drop per eye once daily | ≥2 years: One drop per eye once daily |
| Bepotastine 1.5% (Bepreve, generics) | One drop per eye twice daily | ≥2 years: One drop per eye twice daily |
| Cetirizine 0.24% (Zerviate) | One drop per eye twice daily | ≥2 years: One drop per eye twice daily |
| Epinastine 0.05% (generics) | One drop per eye twice daily | ≥2 years: One drop per eye twice daily |
| Ketotifen fumarate 0.035% (ketotifen 0.025%; multiple OTC products) | One drop per eye twice daily | ≥3 years: One drop per eye twice daily |
| Azelastine 0.05% (generics) | One drop per eye twice daily | ≥3 years: One drop per eye twice daily |
| Emedastine 0.05% (not available in US; available in some other areas) | One drop per eye up to 4 times daily | ≥3 years: One drop per eye up to 4 times daily |
| **Vasoconstrictor/antihistamine combination: Increased redness can occur temporarily when medication is stopped** | | |
| Naphazoline 0.25% and pheniramine 0.3% (OTC Naphcon-A, OTC Visine, OTC Visine-A)<br>Naphazoline 0.27% and pheniramine 0.315% (OTC Opcon-A, generics) | One to two drops per eye up to 4 times daily | ≥6 years: One to two drops per eye up to 4 times daily |
| **Mast cell stabilizers: Decreased itching may be evident within a few days or may take up to four weeks** | | |
| Cromolyn sodium 4% (generics) | One to two drops per eye up to 6 times daily | ≥4 years: One to two drops per eye up to 6 times daily |
| Nedocromil 2% (Alocril) | One to two drops per eye twice daily | ≥3 years: One to two drops per eye twice daily |
| Lodoxamide 0.1% (Alomide) (not available in US; available in some other areas) | One to two drops per eye 4 times daily for up to 3 months | >2 years: One to two drops per eye 4 times daily for up to 3 months |
| Pemirolast 0.1% (Alamast) (not available in US; available in some other areas) | One to two drops per eye up to 4 times daily for up to 4 weeks | ≥3 years: One to two drops per eye up to 4 times daily for up to 4 weeks |

United States trade names are shown in parentheses following generic name.

OTC: over the counter (available without a prescription).

Graphic 91098 Version 17.0

## Contributor Disclosures

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